How-To Library

Informatica

Using PowerExchange for LDAP to Update Information in Active Directory

© Copyright Informatica LLC , 2022. Informatica LLC. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica LLC. All other company and product names may be trade names or trademarks of their respective owners and/or copyrighted materials of such owners.

Abstract

This article describes how to use PowerExchange for LDAP to update a telephone number in Active Directory.

Supported Versions

- PowerExchange for LDAP 10.2 or later

Table of Contents

Overview. 2
Example. 2
Create a Flat File Source Definition. 3
Import the Target. 4
Create a Mapping to Upsert Telephone Values to Active Directory. 9
Create an LDAP Connection. 9
Create and Run the Workflow. 10

Overview

This article describes an example of how to use PowerExchange for LDAP to connect to V3 compliant LDAP directory servers such as Active Directory from PowerCenter.

The example demonstrates the tasks that you must perform to connect to Active Directory and update an attribute from an external data source to Active Directory.

Example

You want to update the telephoneNumber attribute of *uid=fboulbellout* in Active Directory.

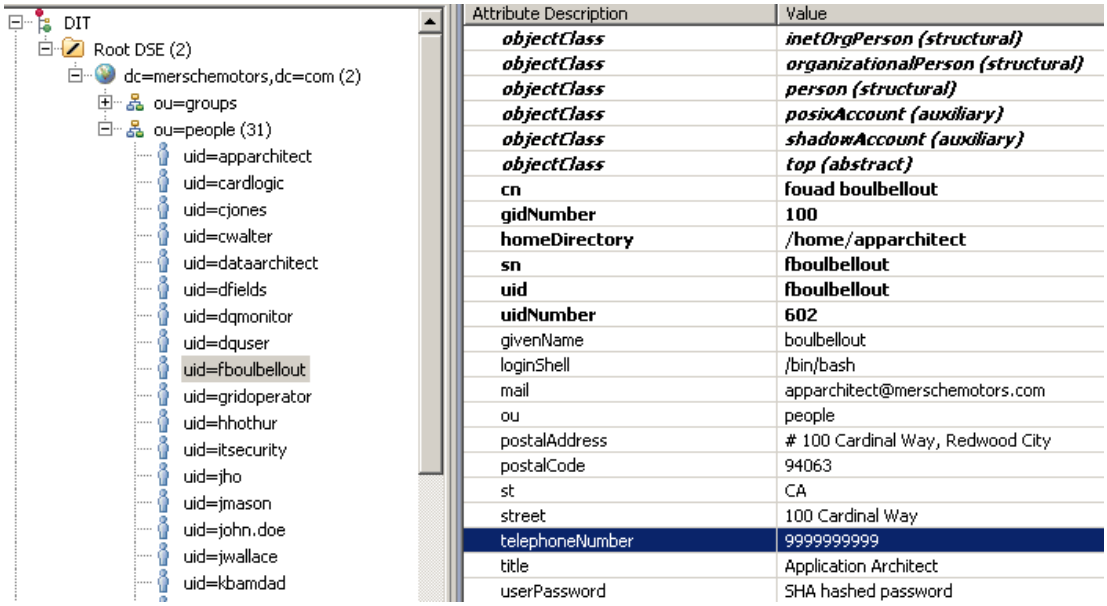
The source file is a CSV file. You want to use PowerExchange for LDAP to integrate the telephone number from the CSV file to Active Directory. The base DN for this attribute is *ou=people,dc=merschemotors,dc=com*.

The setup for this use case includes PowerExchange for LDAP 10.0 configured on PowerCenter 10.0 on a Windows 2012R2 operating system. The OpenLDAP version used is 2.4.44.

The source file contains the following contents that you use for the update:

```
tel;dn;sn;cn
0101010101;uid=fboulbellout,ou=people,dc=merschemotors,dc=com;fboulbellout;fouad boulbellout
```

The following image shows the telephoneNumber attribute that you want to update in Active Directory:



To connect to Active Directory from PowerCenter with PowerExchange for LDAP and update the telephone number in Active Directory, perform the following tasks:

1. Create a flat file source definition to extract the metadata from the flat file.
2. Import the LDAP target definition.
3. Create a mapping and configure an Update Strategy transformation in the mapping to update the telephone values in Active Directory.
4. Create an LDAP connection to connect to Active Directory.
5. Create and run the workflow.

Create a Flat File Source Definition

When you import the source definition, the file must be in a directory local to the client machine. In addition, the PowerCenter Integration Service must be able to access all source files during the session.

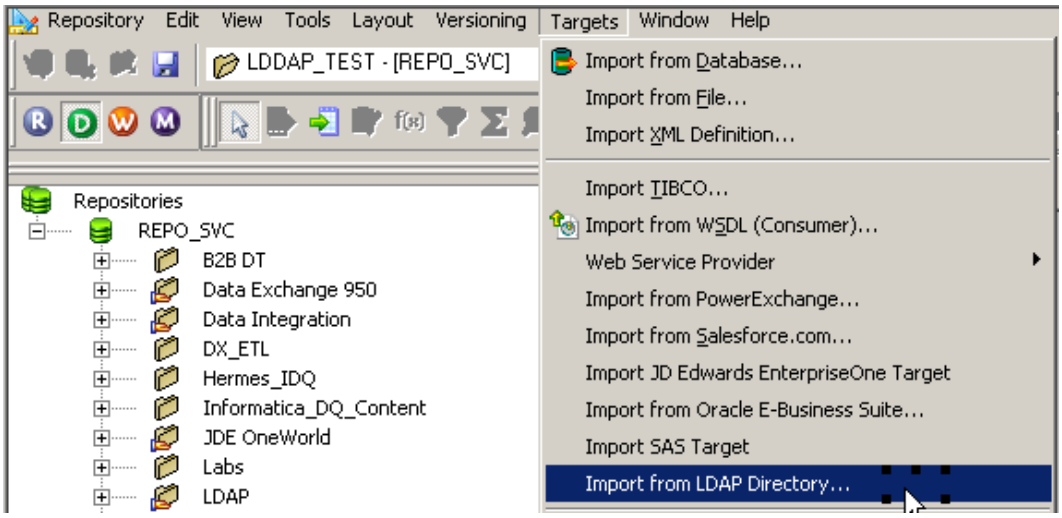
1. Open the Source Analyzer.
2. Click **Sources** > **Import from File**.
3. Select the file that you want to use, and click **OK**.

Import the Target

Import the target definition into the PowerCenter repository. When you import the target definition, the Designer displays a table with columns and LDAP data types.

1. From the **Targets** tab, click **Import from LDAP Directory**.

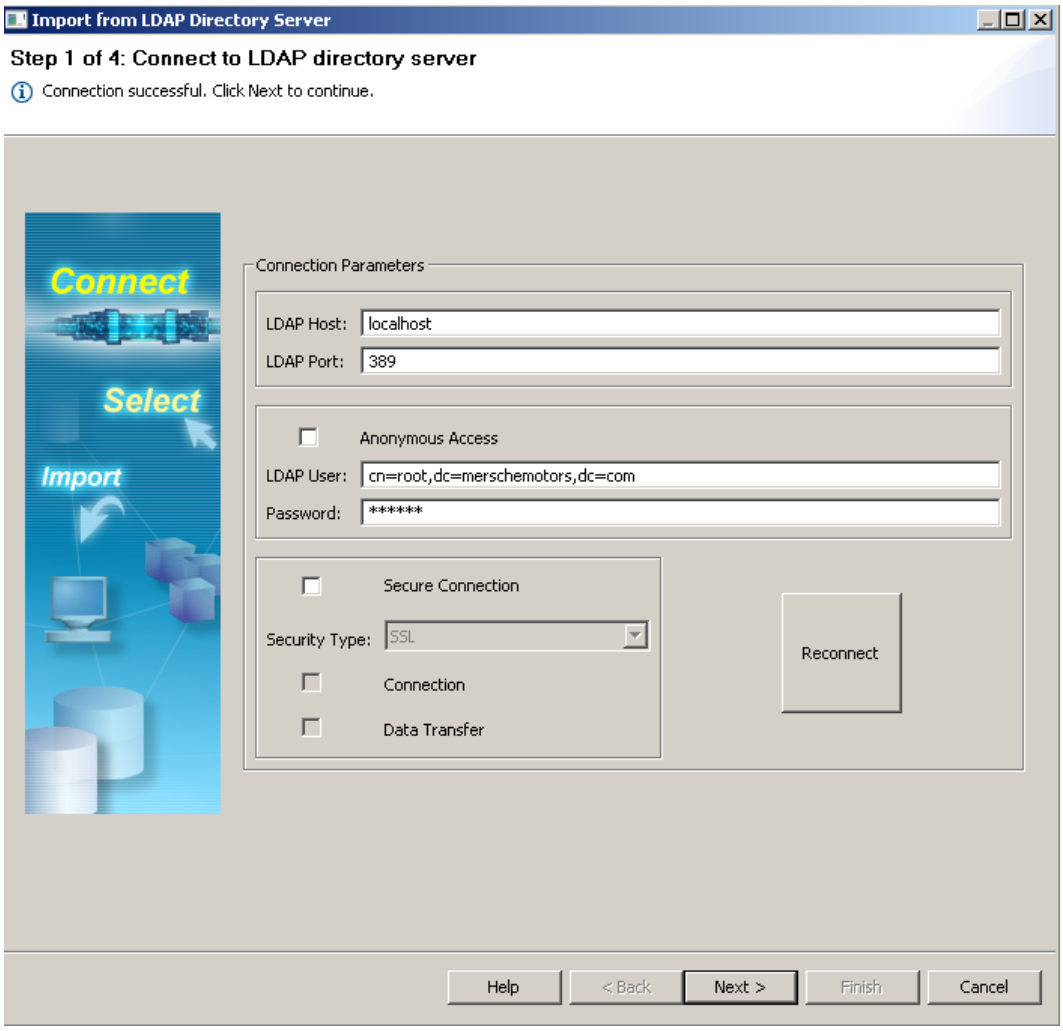
   The following image shows the **Target** tab from which you click the **Import from LDAP Directory** option.



   The Import from LDAP Directory Server wizard appears.

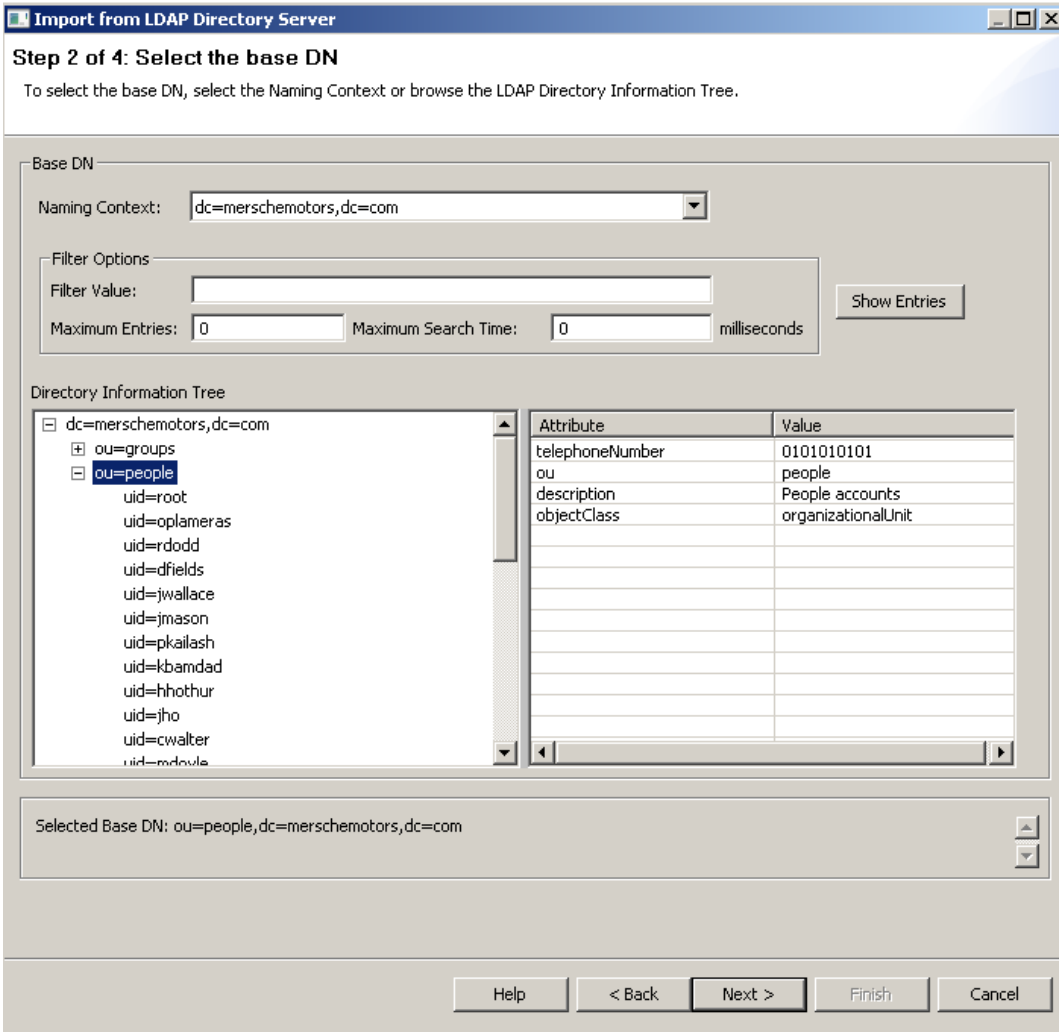
2. Enter the LDAP host, port, user name, and password details.

The following image shows the connection parameters that you must configure:



3. Select the base DN *ou=people,dc=merschemotors,dc=com*, and click **Next**.

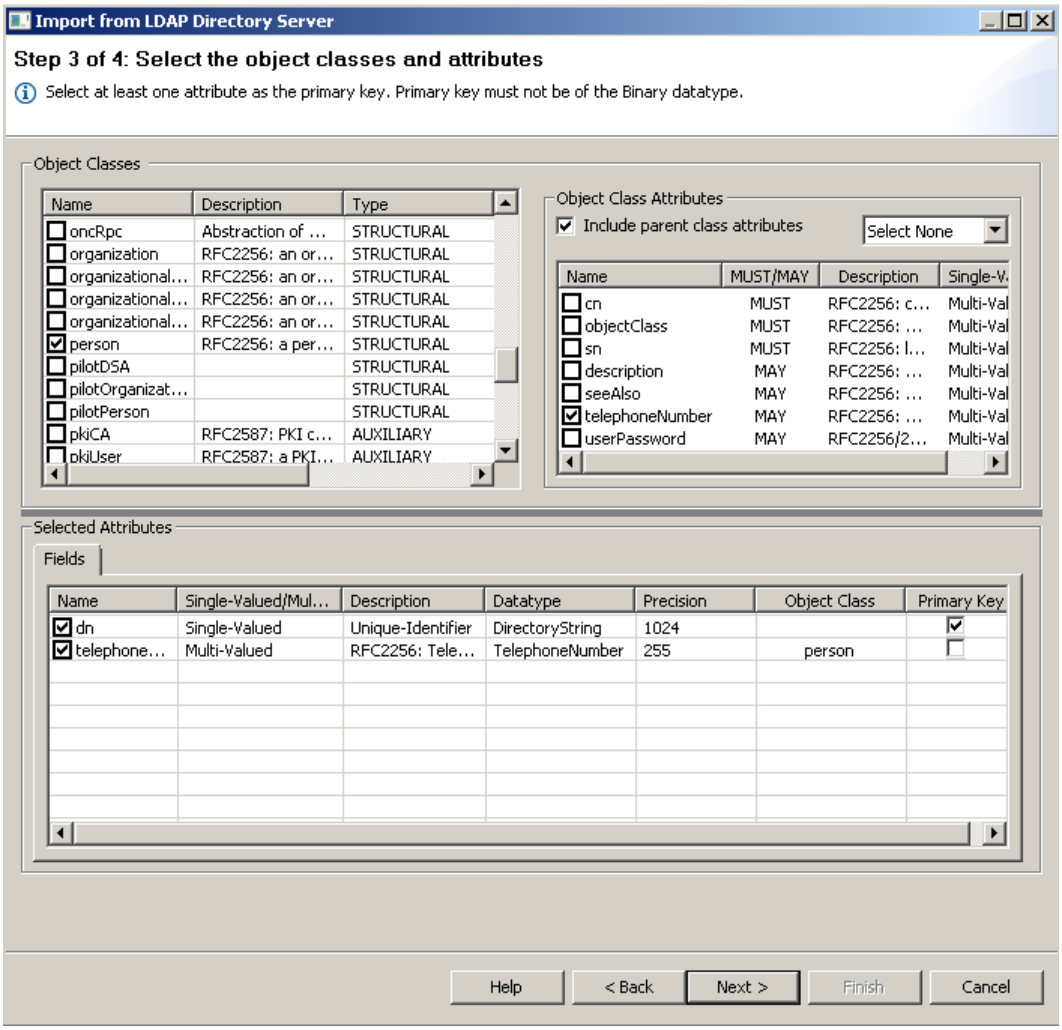
The following image shows the selected base DN from the directory information tree:



Note: The base DN is one level above the information that you want to update in Active Directory.

4. Select DN as the primary key.

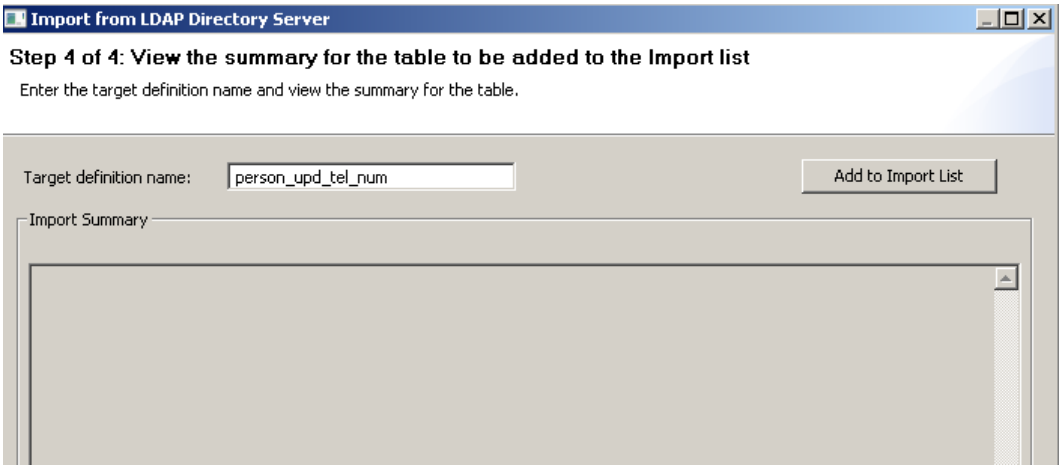
The following image shows the DN attribute selected as the primary key:



Note: As telephoneNumber is a multivalued field, a new group generates.

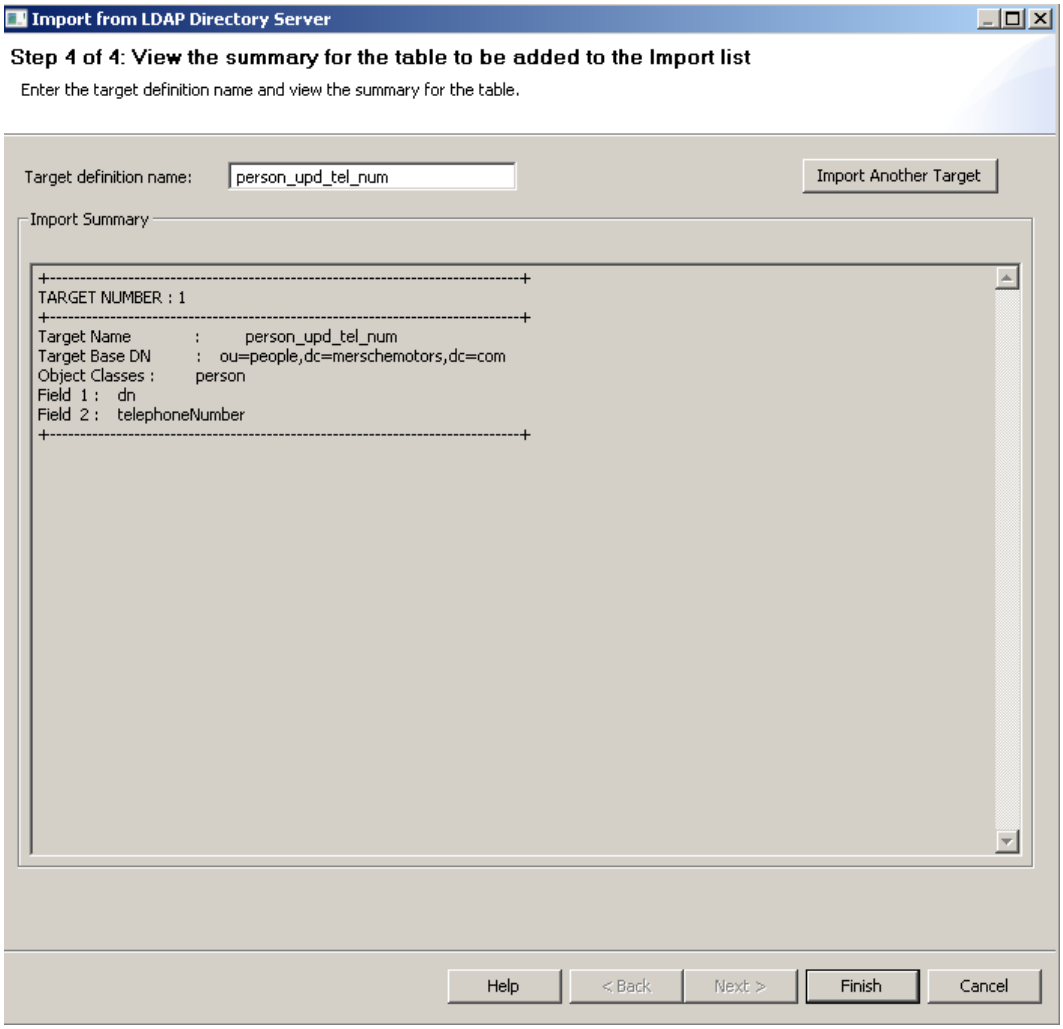
5. Enter a name for the target, and click **Add to Import List**.

   The following image shows the specified target definition name:



6. Verify that the target base DN is correct.
7. Click **Finish**.

   The following image shows the import summary:



   The following image shows the target created with two groups: one for the parent group, and a group each for the multivalued attributes:

person_upd_tel_num (LDAP)

Key Types	Name	Datatype	Length/...
	Parent_Group		
PRIMARY KEY	GPK__Parent_Group	DirectoryString	255
NOT A KEY	Update_Strategy_Parent_Group	Integer	10
PRIMARY KEY	dn	DirectoryString	1024
	telephoneNumber		
PRIMARY KEY	GPK__telephoneNumber	DirectoryString	255
FOREIGN KEY	GFK__Parent_Group__telephoneN...	DirectoryString	255
NOT A KEY	Update_Strategy_telephoneNumber	Integer	10
NOT A KEY	telephoneNumber	TelephoneNumber	255

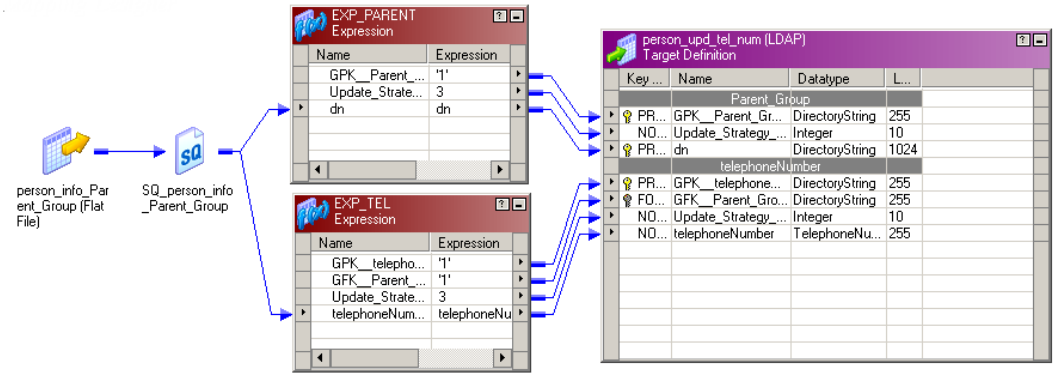
Create a Mapping to Upsert Telephone Values to Active Directory

Create a mapping and then configure an Update Strategy transformation in the mapping to specify the operation that you need to perform on the rows.

When you add an Expression transformation, specify value 3 for upsert in the update strategy field for the parent group. You must hardcode the primary and secondary keys to the same value in the Expression transformation of a mapping because there is only one row. If you have more than one row, you must increment each key and match the keys between the different groups. Each time there is a change in the DN, you must increment the primary key group by one.

1. In the Mapping Designer, create a mapping to upsert the telephone number.
2. Click **Mappings** > **Create**, and provide a name for the mapping.
3. Drag the flat file source and the LDAP target definitions into the mapping.
4. Add the Expression transformation and specify the value for the upsert in the mapping.

   The following image shows the Expression transformation configured in the Mapping Designer:



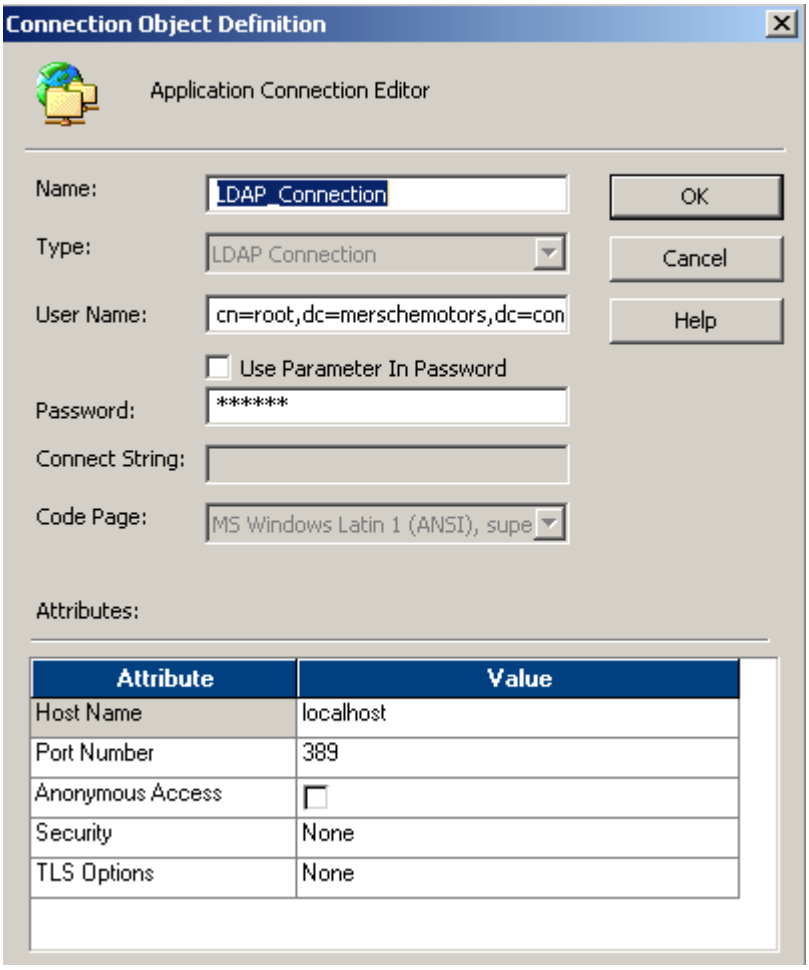
Create an LDAP Connection

Configure an LDAP connection before the PowerCenter Integration Service can write data to the LDAP target. When you configure an LDAP connection, you can specify the connection attributes that the PowerCenter Integration Service uses to connect to the LDAP directory server.

1. In the Workflow Designer, create an LDAP connection.
2. Click **Connections** > **Application**.
3. Click **New**.
4. Select **LDAP Connection** from the **Select Subtype** list.
5. Click **OK**.

6. In the **Connection Object Definition** dialog box, enter the configuration details for the connection object.

   The following image shows the configuration details for the LDAP connection object:



   Note: You cannot establish anonymous access for update, insert, or delete operations.

Create and Run the Workflow

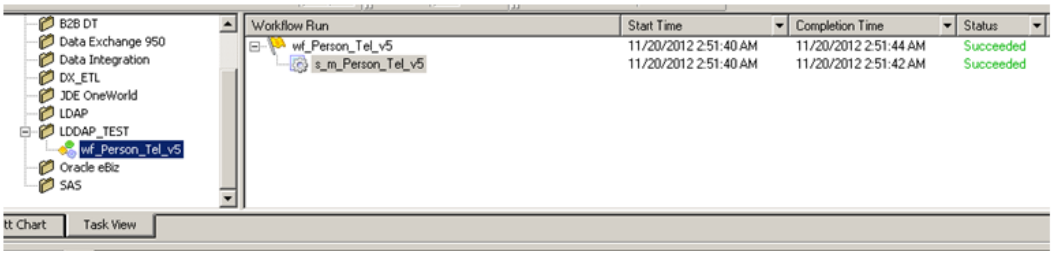
When you create and run a workflow, the PowerCenter Integration Service process starts the process to run the configured session.

1. In the Workflow Designer, click **Workflows** > **Create**.
2. Provide a workflow name, and click **OK**.
3. To add a task, click **Tasks** > **Create**.
4. Select **Session** as the task type.
5. Enter a name for the task, and click **Create**.
6. In the **Mapping** pane, select the created mapping.
7. Click **Tasks** > **Link Task**.
8. Double-click the created session to view the following configured session properties in the mapping:
   - Source. Flat file

- Target. LDAP

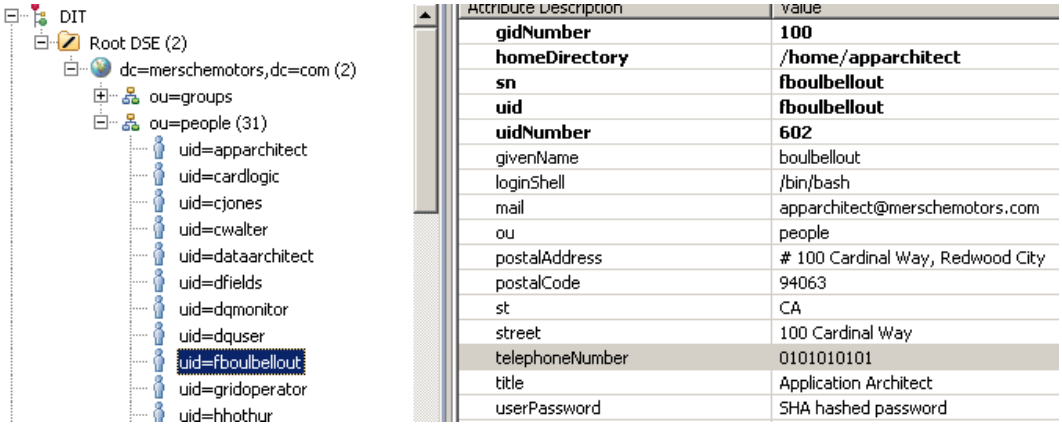
9. Click **Start Workflow**.
10. Verify that there is no error in the session log.

    The following image shows the workflow status:



11. Verify that you can view the data update in Active Directory.

    The following image shows the updated telephone number in Active Directory:



Authors

Dimple Rai

Acknowledgements

The author would like to acknowledge Shiva Kumar Manja for his contribution to this article.